VASANTHAN R

Software Development Engineer II | Backend • Distributed Systems • Agentic AI

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SUMMARY

Backend engineer with **5+ years** at Amazon building durable distributed services, data platforms and AI infrastructure for **100M+ customers** across Prime Video, Kindle and Advertising. A repeat **0-to-1 builder**: led a ground-up rebuild of Prime Video's subtitle quality platform (**97.8% faster, 99.4% cheaper**), built **greenfield** anti-piracy and content-protection systems, a **generative-AI** algorithm framework, and **agentic automation** adopted across the org. Pioneers **AI-native** engineering practices, with agents and **spec-driven** workflows that reshape how the team designs, reviews and ships code.

EXPERIENCE

Software Development Engineer II | Amazon Prime Video & Amazon Advertising **Jun 2024 – Present**

Prime Video: Subtitle Quality Platform (ground-up rebuild)

Tech Stack: Java, Python, AWS (ECS Fargate, Lambda, Step Functions, DynamoDB, SQS, SageMaker), Workflow Orchestration, Agentic AI

- Rebuilt from the **ground up** the platform that checks subtitles in **70+ languages** before they reach **100M+ customers** at **20 TPS**, migrating **50M+ jobs** with zero customer impact via shadow testing and an A/B dial-up.
- Cut **latency 97.8% (3 hours to 8 minutes)** with an event-driven **DAG** workflow engine replacing polling queues, and saved **~$1.3M per year** in ML inference with an idempotent **cache** that skips unchanged videos.
- Pioneered **AI spec-driven development**, where engineers write detailed specs and AI agents write and review the code, cutting feature delivery time from **3 weeks to 1 week** (**47 checks migrated**, zero defects).

Prime Video: Agentic AI for Media Quality

Tech Stack: Python, Amazon Bedrock, Strands, LLMs, Model Context Protocol (MCP), SageMaker

- Designed an **AI perception layer on AWS Strands**, reviewed with a Principal Engineer, that lets AI agents inspect video, audio and subtitles, analyzing each video once and reusing it across **40 languages** (**40x** savings).
- Cut the projected AI **processing cost ~30x** by running agents only on the ~3% of flagged assets, batching signals into one LLM call (**$0.20 to $0.05 per call**) and caching results (**hit rate 5% to 60%**).
- Found that **69% of AI-flagged defects** were creative intent and prototyped a vision and audio agent that separates them at **84% precision**, then cut per-job cost **97% ($0.95 to $0.03)** with a self-hosted open-source LLM.
- Built a shared **agent** toolkit covering 8+ microservices, adopted by 4 squads, that automated on-call handover (manual effort down **95%**, **~1 hour to 1 minute**) and helped resolve **71 production** tickets in 6 months.

Prime Video: Streaming Quality Early-Warning System (greenfield)

Tech Stack: Java, Python, Spark SQL, AWS (EventBridge, Step Functions, ECS Fargate, DynamoDB, OpenSearch, S3)

- Designed and built from **scratch** a system that detects broken viewing experiences on published titles for **100M+ customers**, evaluating **310K** title, territory and language combinations at **100 TPS** through an event-driven pipeline.
- Built a **plug-in architecture** so new data sources and detection rules ship without changing the core engine; exactly-once triggering and **idempotent** writes let failed runs retry safely without duplicate alerts.
- Cut the ingest **latency** from **5 hours to 20 minutes** by moving a **petabyte-scale** data scan off the service's critical path into an upstream batch job that triggers exactly one replayable ingest per run.

Amazon Advertising: Multilingual Contextual Ad Targeting

Tech Stack: Python, Machine Learning, Transformer Models, NLP, Docker, Redis, AWS

- Expanded the system that matches ads to web-page content from English to **8 more languages**, serving **500M+ customers** at **1,500 TPS** and **150M+ daily ad requests**.
- Grew **coverage 68% (3.2M to 5.4M sites per day)** after a data-driven model comparison, recommending a dual-model design (English + multilingual) that preserved English accuracy.
- Cut **pipeline latency from 76 to 30 hours** by publishing predictions directly to **Redis**, and service **startup time 80%** by packaging model files into the container image instead of downloading at boot.

Software Development Engineer I | Amazon Kindle **Jun 2021 – Jun 2024**

Kindle: Distributed Content-Fingerprinting System

Tech Stack: Java, AWS (EC2, ECS, Lambda, DynamoDB, S3, Step Functions), Microservices

- Owned one microservice in a distributed system that invisibly watermarks every e-book download within a **100 ms latency budget** for **250M+ customers** at **5 TPS**; **94M+ deliveries** fingerprinted and **500+ pirates traced**.
- Raised watermark **survivability from 40% to 100%** by tuning image quality for OCR resilience and adding a hidden marker that defeats format-conversion removal, first launched on 100 self-published titles.
- Cut **database cost 40%** and raised **throughput 75%** by re-sharding a hot DynamoDB table, and cut **validation runs 80%** and **failures 72%** by dropping a check that never caught a real regression.

Kindle: Content Protection After a Piracy Breach

Tech Stack: C++, Java, Security, Cryptography, Anti-Tamper, Obfuscation, A/B Testing

- **Led** the engineering response to a Kindle piracy breach, re-securing the apps on Android, iOS, Fire OS and e-readers within a **2-week deadline**, including a new content-encryption scheme.
- Cut anti-tamper performance **overhead from 1200% to 8%** with a **divide-and-conquer** approach across multiple build iterations that isolated the few hot code paths and applied lighter protection only to them.
- Blocked vulnerable app versions server-side through an **A/B dial-up** after a data-driven analysis of 6 mitigation options, and flagged **~5,500 compromised devices** that attached debuggers or tampered with app code.

Kindle: Generative-AI Framework for Encryption Algorithms

Tech Stack: Amazon Bedrock, LLMs, LangChain, Prompt Engineering, C++, Java, AWS Lambda

- Built a **generative AI framework** that auto-generates, validates and benchmarks new encryption algorithms for speed and memory, with automatic C++ and Java translation, shipped to **500K+ devices**.
- Cut new-algorithm development from **~2 months to under 30 days** with an automated compile, test and self-repair loop exposed as a serverless API; design reviewed with a Principal Engineer.
- Built a **70K+ algorithm repository** blending human-written and AI-generated code, with a **weighted-scoring selector** that picks the best candidate for each **latency and memory budget**.

Software Development Engineer Intern | Amazon Kindle **May 2020 – Jul 2020**

- Evaluated and **benchmarked** C/C++ code-**obfuscation** tools on CPU, binary size and cross-platform compatibility to protect Kindle's cryptographic code against reverse engineering, and recommended the production approach.

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, C/C++, Go, TypeScript, JavaScript, Node.js, SQL, Spring, React.js, REST APIs

Cloud & DevOps: AWS, Infrastructure as Code (CDK, Terraform), Docker, Kubernetes, CI/CD, Git

AWS & Data: ECS/Fargate, EC2, Lambda, API Gateway, Step Functions, EventBridge, SQS, SNS, Kafka, DynamoDB, S3, Redis, OpenSearch, Athena, Spark SQL, SageMaker, Bedrock, EMR

Distributed Systems: Large-scale system design, Microservices, Workflow orchestration, CQRS, A/B testing & feature flags, High availability, Fault tolerance, Scalability, Caching, Rate limiting, Idempotency, Sharding, Auto-scaling, Message queues

AI/ML: Agentic AI, LLMs, RAG, Vector databases, Model Context Protocol (MCP), Strands, LangChain, LangGraph, LLM-as-a-judge, Evaluation harnesses, Prompt engineering, Human-in-the-loop, NLP

Engineering Practices: Performance tuning & profiling, Load/performance testing, Unit & Integration testing, TDD, Metric & Alarm observability, Engineering excellence, On-call & Operational excellence, Design & Code reviews, Agile/Scrum

EDUCATION

B.Tech, Information Technology, College of Engineering Guindy, Anna University **2017 – 2021 | CGPA 9.39/10**

ACHIEVEMENTS

- **Runner-up (2nd of 6,700 teams)**, SAP Labs India Online Hackathon: built a solution for "Bringing **Explainable AI** into Business Applications with SAP".